

XINHAI CHANG

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RESEARCH INTERESTS

My research centers on closing the **Real-Sim-Real** loop for robot learning: capturing real-world environments into high-fidelity digital twins, training generalizable policies at scale in simulation, and deploying back to the physical world. My interests span **3D vision** (reconstruction, generation, and scene understanding) and **embodied AI** (simulation, world models, mobile manipulation, and sim-to-real transfer).

EDUCATION

Bachelor of Science in Data Science, Peking University Expected June 2027
Yuanpei College, Beijing, China
GPA: 3.76/4.0
Coursework: Computer Vision, Multimodal Learning, Intelligent Robotics [3.9+]

RESEARCH EXPERIENCE

Student Researcher, VCL Lab, Peking University June 2026 – Present
Advised by Prof. Wenzheng Chen

- Research on robot learning and world models for embodied AI, with an emphasis on depth perception, navigation, and real-world robotic manipulation.

Student Researcher, Media Lab and EECS, MIT March 2024 – Present
Mentored by Dr. Kaichen Zhou

- Research on 3D/4D perception and generative world models, including 3D reconstruction, 3D editing, 3D generation, world action models, and multi-view benchmarks.

Student Researcher, CoRe Lab, Peking University March 2025 – June 2026
Advised by Prof. Yixin Zhu

- Research on scalable simulation and planning for mobile and dexterous manipulation, including trajectory generation, tactile-guided control, and policy evaluation.

TECHNICAL SKILLS

Robotics: Robot manipulation, mobile manipulation, dexterous manipulation, trajectory planning, large-scale simulation, visuomotor policy learning and evaluation; Isaac Lab, Isaac Sim, cuRobo.

3D Vision: 3D reconstruction, 3D editing, 3D generation, 4D perception, world models, SLAM, depth estimation, multimodal learning.

Programming: Python, C/C++, Bash, HTML, CSS; PyTorch, TensorFlow, Pandas, Qt.

PUBLICATIONS

* equal contribution † corresponding author ● computer vision ● robotics

- [7] Jiaheng Li, Binsheng Zhang, **Xinhai Chang**, Wenzheng Chen[†]. *NSL-SLAM: High-Fidelity Neural Structured-Light Depth for Practical SLAM and Reconstruction*. ACM International Conference on Multimedia (ACM MM), 2026. ●
- [6] Kaichen Zhou*, Yuzhen Chen*, Fangneng Zhan, Hang Hua, Grace Chen, **Xinhai Chang**, Ao Qu, Yilun Du, Zhuang Liu, Paul Pu Liang[†], Mengyu Wang[†]. *GEM-4D: Geometry-Enhanced Video World Models for Robot Manipulation*. European Conference on Computer Vision (ECCV), 2026. CVPR WMAS 2026 Best Paper Finalist. ● ●
- [5] Yida Niu*, **Xinhai Chang***, Xin Liu*, Ziyuan Jiao[†], Yixin Zhu[†]. *Scalable Trajectory Generation for Whole-Body Mobile Manipulation*. Computer Vision and Pattern Recognition (CVPR), 2026. Highlight. ●
- [4] **Xinhai Chang**, Kaichen Zhou[†]. *Neural Surface Reconstruction from Sparse Views Using Epipolar Geometry*. Computer Vision and Pattern Recognition Workshop A2AMML, 2026. ●
- [3] Kaichen Zhou, Yuhan Wang*, Grace Chen*, **Xinhai Chang***, Gaspard Beaudouin, Fangneng Zhan, Paul Pu Liang[†], Mengyu Wang[†]. *PAGE-4D: Disentangled Pose and Geometry Estimation for 4D Perception*. International Conference on Learning Representations (ICLR), 2026. ●

- [2] Hongze Li*, Xuchuan Huang*, **Xinhai Chang***, Jun Zhou, Huijing Zhao[†]. *TerraX: Visual Terrain Classification Enhanced by Vision-Language Models*. International Conference on Intelligent Robots and Systems (IROS), Oral, 2025. ● ●
- [1] Kaichen Zhou*, Lanqing Hong*, **Xinhai Chang***, Yingji Zhong, Enze Xie, Hao Dong, Zhihao Li, Yongxin Yang, Zhenguo Li, Wei Zhang. *SplatMesh: Interactive 3D Segmentation and Editing Using Mesh-Based Gaussian Splatting*. CVPR 2025 Workshop. ●

PREPRINTS

* equal contribution † corresponding author ● computer vision ● robotics

- [2] Kaichen Zhou*, **Xinhai Chang***, Taewhan Kim*, Jiadong Zhang*, Yang Cao, Chufei Peng, Fangneng Zhan, Hao Zhao, Hao Dong, Kai Ming Ting, Ye Zhu[†]. *RAD: A Realistic Multi-View Benchmark for Pose-Agnostic Anomaly Detection*. Under Review, 2026. ● ●
- [1] Kaichen Zhou*, Zeyang Bai*, **Xinhai Chang***, Mengyu Wang[†], Paul Pu Liang[†], Fangneng Zhan[†]. *Stream3D: Sequential Multi-View 3D Generation via Evidential Memory*. Under Review, 2026. ECCV OpenSUN3D 2026 Best Paper Finalist. ●

HONORS AND AWARDS

May Fourth Medal (Peking University's Highest Student Honor)	2026
Best Paper Finalist, ECCV 2026 OpenSUN3D Workshop (Stream3D)	2026
Best Paper Finalist, CVPR 2026 WMAS Workshop (GEM-4D, \$1500)	2026
National Scholarship (China's Highest Honor for Undergraduate Students)	2025
Second-class Scholarship, Peking University	2024
First Prize, 21st Jiang Zehan Cup Mathematical Modeling Competition	2024
Xing Zhengde Scholarship, Yuanpei College	2023
Third-class Scholarship, Peking University	2023

TEACHING EXPERIENCE

Teaching Assistant , Introduction to Computation B (Prof. Jun Sun)	Autumn 2025
Led weekly programming labs, graded assignments, and designed exam questions.	
Teaching Assistant , Introduction to Computation C (Prof. Baobao Chang)	Autumn 2025
Led weekly programming labs, graded assignments, and designed exams and the final project.	
Teaching Assistant , Introduction to Computer Systems (Prof. Lu Zhang)	Autumn 2024
Led weekly discussion sessions, graded assignments, and contributed to exam design.	

SERVICE AND OUTREACH

Organizer , NeurIPS 2026 Workshop: Physical World Model	2026
Volunteer Teacher, Student Development Center, Peking University	Spring 2026
Volunteer Teacher, Student Development Center, Peking University	Autumn 2025
Volunteer Teacher, International Asian Liver Center, Gansu, China	May – Aug. 2024
Volunteer, Yunnan Golden Monkey Conservation, Peking University	Winter 2023